

EEE4114F: Problem set 2 solutions

1. Note that we may write $x(n)$ as the sum of a left-sided sequence and a right-sided sequence as follows:

$$x(n) = \left(\frac{1}{4}\right)^n u(n) + \left(\frac{1}{4}\right)^{-n} u(-n) - \delta(n)$$

where the last term is included to remove the extra term that is introduced at $n = 0$ by the two exponential sequences. The DTFT of the first term is

$$\left(\frac{1}{4}\right)^n u(n) \Leftarrow DTFT \Rightarrow \frac{1}{1 - \frac{1}{4}e^{-j\omega}}$$

and, using the time reversal property, it follows that the DTFT of the second term is

$$\left(\frac{1}{4}\right)^{-n} u(-n) \Leftarrow DTFT \Rightarrow \frac{1}{1 - \frac{1}{4}e^{j\omega}}$$

Therefore,

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{1 - \frac{1}{4}e^{-j\omega}} + \frac{1}{1 - \frac{1}{4}e^{j\omega}} - 1 \\ &= \frac{\frac{15}{16}}{(1 - \frac{1}{4}e^{-j\omega})(1 - \frac{1}{4}e^{j\omega})} = \frac{\frac{15}{16}}{\frac{17}{16} - \frac{1}{2}\cos\omega} \end{aligned}$$

2. a. This system has a unit sample response that is finite in length. Therefore, the frequency response is a polynomial in $e^{j\omega}$, with the coefficients of the polynomial equal to the values of $h(n)$:

$$H(e^{j\omega}) = 1 + 6e^{-j\omega} + 3e^{-2j\omega}$$

This can be written more formally as

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} h(n)e^{-jn\omega} \\ &= \sum_{n=-\infty}^{\infty} [\delta(n) + 6\delta(n-1) + 3\delta(n-2)]e^{-jn\omega} \end{aligned}$$

Because

$$\sum_{n=-\infty}^{\infty} \delta(n - n_0)e^{-jn\omega} = e^{-jn_0\omega}$$

then

$$H(e^{j\omega}) = 1 + 6e^{-j\omega} + 3e^{-2j\omega}$$

b. For the second system, the frequency response is

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-jn\omega} = \sum_{n=2}^{\infty} \left(\frac{1}{3}\right)^{n+2} e^{-jn\omega}$$

Changing the limits on the sum so that it begins with $n = 0$, we have

$$H(e^{j\omega}) = \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^{n+4} e^{-j(n+2)\omega} = \left(\frac{1}{3}\right)^4 e^{-2j\omega} \sum_{n=0}^{\infty} \left(\frac{1}{3}e^{-j\omega}\right)^n$$

Using the series, we find

$$H(e^{j\omega}) = \left(\frac{1}{3}\right)^4 \frac{e^{-2j\omega}}{1 - \frac{1}{3}e^{-j\omega}}$$

3. This problem may be solved using the eigenfunction property of LSI systems. If the input to an LSI system is $x(n) = \cos(n\omega_0)$, the response will be

$$y(n) = |H(e^{j\omega_0})| \cos(n\omega_0 + \phi_h(\omega_0))$$

We need to find the frequency response of the system. The unit sample response of an ideal LPF,

$$H_1(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq \omega_c \\ 0 & \omega_c < |\omega| \leq \pi \end{cases}$$

is

$$h_1(n) = \frac{\sin(n\omega_c)}{\pi n}$$

Because $h(n) = 2h_1(n-1)$ with $\omega_c = \pi/2$, an expression may be derived for $H(e^{j\omega})$ in terms of $H_1(e^{j\omega})$ as follows:

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} h(n)e^{-jn\omega} = \sum_{n=-\infty}^{\infty} 2h_1(n-1)e^{-jn\omega} \\ &= 2 \sum_{n=-\infty}^{\infty} h_1(n)e^{-j(n+1)\omega} = 2e^{-j\omega} \sum_{n=-\infty}^{\infty} h_1(n)e^{-jn\omega} = 2e^{-j\omega} H_1(e^{j\omega}) \end{aligned}$$

Therefore,

$$H(e^{j\omega}) = \begin{cases} 2e^{-j\omega} & |\omega| \leq \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\omega| \leq \pi \end{cases}$$

Because $|H(e^{j\omega})| = 0$ at $\omega = 3\pi/4$, the sinusoid in $x(n)$ is filtered out, and the output is simply

$$y(n) = 2|H(e^{j\pi/4})| \cos\left(\frac{n\pi}{4} + \phi_h\left(\frac{\pi}{4}\right)\right)$$

$$y(n) = 4 \cos(n\pi/4 + \pi/4)$$

4. a. The unit sample response may be found in two different ways. The first is to use the inverse DTFT formula and perform the integration. The second is to use the modulation property and note that if

$$H_{lp}(e^{j\omega}) = \begin{cases} 1 & \text{for } |\omega| \leq \pi/4 \\ 0 & \text{otherwise} \end{cases}$$

$H(e^{j\omega})$ may be written as

$$H(e^{j\omega}) = H_{lp}(e^{j(\omega-\pi)})$$

Therefore, it follows from the modulation property that

$$h(n) = e^{jn\pi} h_{lp}(n) = (-1)^n h_{lp}(n)$$

With

$$h_{lp}(n) = \frac{\sin(n\pi/4)}{n\pi}$$

we have

$$h(n) = (-1)^n \frac{\sin(n\pi/4)}{n\pi}$$

- b. The frequency response of the system that has a unit sample response $h_1(n) = h(2n)$ may be found by evaluating the discrete-time Fourier transform sum directly:

$$H_1(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h_1(n) e^{-jn\omega} = \sum_{n=-\infty}^{\infty} h(2n) e^{-jn\omega} = \sum_{n \text{ even}} h(n) e^{-jn\omega/2}$$

However, an easier approach is to note that

$$h_1(n) = h(2n) = (-1)^{2n} \frac{\sin(2n\pi/4)}{2n\pi} = \frac{\sin(n\pi/2)}{2n\pi}$$

which is a low-pass filter with a cutoff frequency of $\pi/2$ and a gain of 1/2. A plot of $H_1(e^{j\omega})$ is shown in the following figure:

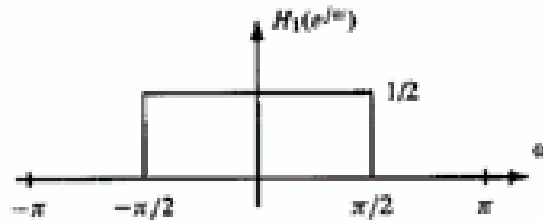


Figure 7

5. To find the unit sample response, let $x(n) = \delta(n)$. The output of this adder is then

$$\omega(n) = \delta(n) - \delta(n-1)$$

Because $\omega(n)$ is input to an LSI system with a unit sample response $h_2(n)$,

$$y(n) = h_2(n) - h_2(n-1)$$

Where

$$h_2(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_2(e^{j\omega}) e^{jn\omega} d\omega = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{jn\omega} d\omega = \frac{\sin(n\pi/2)}{n\pi}$$

Therefore, the unit sample response of the overall system is

$$h(n) = \frac{\sin(n\pi/2)}{n\pi} - \frac{\sin[(n-1)\pi/2]}{(n-1)\pi}$$

To find the frequency response, note that

$$W(e^{j\omega}) = 1 - e^{-j\omega}$$

Therefore,

$$\begin{aligned} H(e^{j\omega}) &= W(e^{j\omega})H_2(e^{j\omega}) = [1 - e^{-j\omega}]H_2(e^{j\omega}) \\ &= \begin{cases} 1 - e^{-j\omega} & |\omega| \leq \pi/2 \\ 0 & \pi/2 < |\omega| \leq \pi. \end{cases} \end{aligned}$$

6. a. For the first sequence, the DTFT may be evaluated directly as follows:

$$\begin{aligned} X_1(e^{j\omega}) &= \sum_{n=-3}^{\infty} \left(\frac{1}{2}\right)^n e^{-jn\omega} = \sum_{n=-3}^{\infty} \left(\frac{1}{2}e^{-j\omega}\right)^n \\ &= \left(\frac{1}{2}e^{-j\omega}\right)^{-3} \sum_{n=0}^{\infty} \left(\frac{1}{2}e^{-j\omega}\right)^n = \frac{8e^{3j\omega}}{1 - \frac{1}{2}e^{-j\omega}} \end{aligned}$$

b. The best way to find the DTFT of $x_2(n)$ is to express the sinusoid as a sum of two complex exponentials as follows:

$$x_2(n) = \frac{1}{2j} [\alpha^n e^{jn\omega_0} - \alpha^n e^{-jn\omega_0}] u(n)$$

The DTFT of the first term is

$$\frac{1}{2j} \sum_{n=0}^{\infty} \alpha^n e^{jn\omega_0} e^{-jn\omega} = \frac{1}{2j} \sum_{n=0}^{\infty} \left(\alpha e^{-j(\omega-\omega_0)}\right)^n = \frac{1}{2j} \frac{1}{1 - \alpha e^{-j(\omega-\omega_0)}}$$

Similarly, for the second term we have

$$-\frac{1}{2j} \sum_{n=0}^{\infty} \alpha^n e^{-jn\omega_0} e^{-jn\omega} = -\frac{1}{2j} \frac{1}{1 - \alpha e^{-j(\omega+\omega_0)}}$$

Therefore,

$$\begin{aligned} X_2(e^{j\omega}) &= \frac{1}{2j} \left[\frac{1}{1 - \alpha e^{-j(\omega-\omega_0)}} - \frac{1}{1 - \alpha e^{-j(\omega+\omega_0)}} \right] \\ &= \frac{(\alpha \sin \omega_0) e^{-j\omega}}{1 - (2\alpha \cos \omega_0) e^{-j\omega} + \alpha^2 e^{-2j\omega}} \end{aligned}$$

c. Finally, for $x_3(n)$, we have

$$X_3(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x_3(n) e^{-jn\omega} = \sum_{n=0,2,4,\dots}^{\infty} \left(\frac{1}{2}\right)^n e^{-jn\omega}$$

Therefore

$$X_3(e^{j\omega}) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{2n} e^{-2jn\omega} = \sum_{n=0}^{\infty} \left(\frac{1}{4} e^{-2j\omega}\right)^n = \frac{1}{1 - 1/4 e^{-2j\omega}}$$

17. a. The DTFT of $x^*(-n)$ is

$$\text{DTFT}\{x^*(-n)\} = \sum_{n=-\infty}^{\infty} x^*(-n) e^{-jn\omega} = \sum_{n=-\infty}^{\infty} x^*(n) e^{jn\omega}$$

Bringing the conjugate outside, we have

$$\text{DTFT}\{x^*(-n)\} = \left[\sum_{n=-\infty}^{\infty} x(n) e^{-jn\omega} \right]^* = X^*(e^{j\omega})$$

which leads to the DTFT pair

$$x^*(-n) \Leftarrow DTFT \Rightarrow X^*(e^{j\omega})$$

b. For $y(n) = x(n) * x^*(-n)$, note that because $y(n)$ is the convolution of the two sequences, the DTFT of $y(n)$ is the product of the DTFTs of $x(n)$ and $x^*(-n)$. As shown in part (a), the DTFT of x^* is $X^*(e^{j\omega})$. Therefore, we have the DTFT pair

$$x(n) * x^*(-n) \Leftarrow DTFT \Rightarrow X(e^{j\omega}) X^*(e^{j\omega}) = |X(e^{j\omega})|^2$$

c. For $x(2n+1)$ we have

$$\text{DTFT}\{x(2n+1)\} = \sum_{n=-\infty}^{\infty} x(2n+1) e^{-jn\omega} = \sum_{n \text{ odd}} a(n) e^{-jn\omega}$$

To evaluate this sum, a "trick" is to use the identity

$$1 - (-1)^n = \begin{cases} 2 & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

This allows us to write the DTFT as follows:

$$\begin{aligned} \text{DTFT}\{x(2n+1)\} &= \sum_{n \text{ odd}} x(n)e^{-jn\omega} = \frac{1}{2} \sum_{n=-\infty}^{\infty} [1 - (-1)^n]x(n)e^{-jn\omega} \\ &= \frac{1}{2} \sum_{n=-\infty}^{\infty} x(n)e^{-jn\omega} - \frac{1}{2} \sum_{n=-\infty}^{\infty} (-1)^n x(n)e^{-jn\omega} \end{aligned}$$

Because the first sum is simply $X(e^{j\omega})$, and the second DTFT of the modulated signal $(-1)^n x(n) = e^{jn\pi} x(n)$ then

$$\text{DTFT}\{x(2n+1)\} = 1/2[X(e^{j\omega}) - X(e^{j(\omega-\pi)})]$$

8. To find the frequency response, we begin by finding the DTFT of each term in the difference equation

$$(1 - 0.5e^{-j\omega})Y(e^{j\omega}) = (1 + 2e^{-j\omega} = e^{-j2\omega})X(e^{j\omega})$$

Because $H(e^{j\omega}) = Y(e^{j\omega})/X(e^{j\omega})$, we have

$$H(e^{j\omega}) = \frac{1 + 2e^{-j\omega} + e^{-2j\omega}}{1 - 0.5e^{-j\omega}}.$$